



SSR RADAR

During earlier testing, it was observed that specific microwave Doppler radar modules available on the market provide insufficient output current to trigger the optotriac in some cases, resulting in difficulty for the optocouplers when switching an AC bulb. To address this behaviour, a solder jumper was included to allow connection to a development board, enabling radar data processing and reliable bulb activation through the board. A buffer circuit or transistor may also be added if such issues arise with a specific radar module.

For enhanced reliability and compatibility with all RCWL microwave radar modules on the market, a buffer transistor circuit is recommended when direct triggering is not feasible. In this system, the radar output is connected directly to the development board, as testing confirmed reliable performance without a transistor buffer. Fig. 1 illustrates the radar UI dashboard and the radar PCB design.

Circuit diagram

The radar module's output is connected to IO pin-1 on the board via a solder jumper. The optocoupler input pin is connected via a solder jumper to IO pin-3 for control and to pin-4 to read the light's on/off state. The entire DC section is powered by a 3.3V AC-DC converter module that drives the DC circuitry.